MASTER PROMPT – Semera Logiya Building Permit Telegram Bot

You are a senior Python developer, Telegram Bot expert, FastAPI backend engineer, PostgreSQL database architect, and government digital transformation consultant.

Build a **production-ready Telegram Bot** for the **Semera Logiya Building Permit Office** in Ethiopia.

The objective is to digitize the complete building permit application, design review, approval, revision, and permit issuance process.

The system must include a Telegram Bot for applicants, a web-based dashboard for engineers and administrators, secure file management, application tracking, notifications, and role-based access control.

Workflow

```
                      SEMERA LOGIYA BUILDING PERMIT WORKFLOW

Citizen
    │
    ▼
Open Telegram Bot
    │
    ├───────────────────────────────────────────────┐
    │                                               │
    ▼                                               ▼
New Application                           🔍 Check / Find Application
    │                                               │
    ▼                                               ▼
Register                                  Enter One of the Following:
(Full Name + Phone Number)                • Application ID
    │                                     • National ID
    ▼                                     • Full Name + Phone Number
Submit New Building Permit Application              │
    │                                               ▼
    ▼                                      Bot Searches Database
Upload Required Files                               │
• DWG Drawing                                       ▼
• PDF Drawing                              Application Found?
    │                                     ┌───────────┴──────────┐
    ▼                                     │                      │
```

```
Bot Generates Application ID              ▼                     ▼
                                          Yes                   No
SLBP-2026-000125                          │                     │
    │                                     ▼                     ▼
    ▼                             Display Current Status   No Application Found
Application Submitted Successfully         │
    │                                     ▼
    ▼                             Status Examples:
Status:                            🟡 Pending Design Review
🟡 Pending Design Review             🔄 Revision Required
    │                              🟢 Design Approved
    ▼                              ✅ Permit Issued
Bot Automatically Sends Files               │
to Design Review Department                 ▼
    │                              If Revision Required
    ▼                              Show Engineer Comments
Design Engineer Reviews Drawings
    │
    ▼
Decision
──────────────────────────────────────────────────────────────
│                                                            │
│                                                            │
▼                                                            ▼
✅ Approved                                           ❌ Revision Required /
Rejected
│                                                            │
│                                                            ▼
│                                                    Engineer Adds Comments
│                                                    Example:
│                                                    • Increase staircase width
│                                                    • Update foundation detail
│                                                    • Correct column dimensions
│                                                    • Missing electrical layout
│                                                    • Add structural
calculations
│                                                            │
│                                                            ▼
│                                                  Bot Sends Comments to Applicant
│                                                            │
│                                                            ▼
│                                                  Applicant Updates the Design
│                                                            │
│                                                            ▼
│                                                  Upload Revised DWG/PDF Files
│                                                            │
```

```
|                                                        ▼
|                                            Status:
|                                            🔄 Resubmitted for Review
|                                                        |
└───────────────────────────────────────◄────────────────┘
                    |
                    ▼
             Design Approved
                    |
                    ▼
         Bot Notifies Applicant
                    |
                    ▼
         Print Approved Drawings
                    |
                    ▼
Bring the Following Mandatory Documents
                    |
                    ▼
📄 Land Documents
• Original Land Ownership Certificate
• Copy of Land Ownership Certificate
                    |
                    ▼
📐 Design Documents
• Approved Architectural Drawings
• Structural Drawings & Details
• Electrical Drawings
• Mechanical Drawings (if applicable)
• Foundation & Reinforcement Details
• Printed and Digital (PDF/DWG) Drawings
                    |
                    ▼
📊 Construction Documents
• Bill of Quantities (BOQ)
• Structural Calculation Report (if required)
• Material Specifications (if required)
• Technical Reports (if required)
                    |
                    ▼
👤 Applicant Documents
• National ID
• Other Required Documents
                    |
                    ▼
Visit Semera Logiya Building Permit Office
                    |
```

```
                          ▼
Document Verification
                          │
                          ▼
Building Permit Processing
                          │
                          ▼
Final Approval
                          │
                          ▼
🎉 Building Permit Issued
```

Functional Requirements

Applicant

- Register with Full Name and Phone Number.
- Submit a new building permit application.
- Upload DWG and PDF drawings.
- Receive a unique application number.
- Track application status.
- Receive Telegram notifications for every status update.
- View engineer comments if revisions are requested.
- Upload revised drawings.
- Receive approval notifications.
- View the list of mandatory documents after approval.
- Check application status from **any phone** by selecting **Check / Find Application** and searching with:
- Application ID
- National ID
- Full Name + Phone Number

Design Engineer

- Secure login.
- View assigned applications.
- Open DWG/PDF files.
- Download drawings.
- Approve applications.
- Reject applications.
- Request revisions.
- Enter comments explaining required corrections.
- Automatically notify applicants after every action.

Administrator

- Manage users and engineers.
- Assign applications to departments.
- Monitor pending applications.
- View workflow statistics.
- Search applications.
- Export reports to Excel and PDF.
- Configure departments and permissions.

Departments

- Design Review Department
- Building Permit Office
- System Administrator

The system should support adding more departments in the future.

Statuses

- Submitted
- Pending Design Review
- Under Review
- Revision Required
- Resubmitted
- Approved
- Ready for Printing
- Waiting for Documents
- Document Verification
- Building Permit Processing
- Permit Issued
- Rejected

File Upload

Support:

- DWG
- PDF

Maximum file size: 50 MB.

Store all files securely and associate them with the application ID.

Notifications

Automatically notify applicants when:

- Application submitted
- Under review
- Revision requested
- Engineer comments added
- Revised files received
- Approved
- Ready for printing
- Permit issued

Database

Store:

- Applicant details
- Telegram ID
- Full Name
- Phone Number
- National ID
- Application ID
- Submission Date
- Current Status
- Assigned Engineer
- Engineer Comments
- Approval History
- Revision History
- Uploaded Files
- Notification History

Technology Stack

Backend:

- Python

- FastAPI

Telegram Bot:

- aiogram 3.x

Database:

- PostgreSQL

ORM:

- SQLAlchemy

Storage:

- Local Storage or MinIO

Authentication:

- JWT + Telegram User ID

Frontend Dashboard:

- React + Tailwind CSS

Deployment:

- Docker
- Ubuntu Server
- Nginx

UI/UX

Create a modern, intuitive interface with:

- Telegram inline keyboards.
- Progress timeline for each application.
- Search functionality.
- Responsive admin dashboard.
- Dark and light mode.
- Role-based navigation.

Deliverables

Generate a complete, production-ready solution including:

- Telegram Bot source code.
- FastAPI backend.
- PostgreSQL database schema.
- React admin dashboard.
- Engineer dashboard.
- File upload service.
- Authentication and authorization.
- Notification system.
- Docker configuration.
- Environment configuration.
- API documentation.
- Deployment guide.
- README.
- Clean, modular project structure.
- Well-documented and maintainable code.

The application should be secure, scalable, easy to maintain, and suitable for deployment as the official digital building permit system for the Semera Logiya Building Permit Office.